

Review retrospective:
the universal large- T rate — a sound estimate and five dead locals

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Abstract

A soundness review of `Laplace/OneD/UniversalAsymptotics.lean`: the rate-aware large- T asymptotic of the universal scaling function, $|T\langle y^2 \rangle_{V,T} - 1| \leq K/T$ for the potential $V(y) = y^2/2 + y^4/24$. Result: **a clean fidelity pass with six edits** — a dead **open Real** and five dead proof-locals. The substantive estimate is correct and was checked on six independent points.

1 Setting

This is the $T \gg 1$ companion to the universal-scaling identity: where that file gives the exact transport, this one supplies the large- T rate $Tf_1(T) \rightarrow 1$ with explicit $O(1/T)$ bound, by bridging to the seabed's `I_0.asymptotic/I_2.asymptotic` at $(\lambda, \alpha, \gamma) = (1, 0, 1)$ (the discriminant $\alpha^2 < 3\lambda\gamma$ becoming $0 < 3$).

2 What was reviewed

Two public theorems (the rate bound and its **Tendsto** corollary) and a trivial private discriminant lemma.

3 Fidelity / soundness findings

Sound; checked six ways. (1) The leading constant $a = \sqrt{2\pi}$ is right ($\sqrt{T} Z_V \rightarrow \int e^{-z^2/2} = \sqrt{2\pi}$ after $y = z/\sqrt{T}$). (2) The limit is 1, not $1/2$ or 2 ($TN/Z \rightarrow a/a = 1$). (3) The rate is $O(1/T)$ with witness $K = 2(K_0 + K_2)/a$. (4) The denominator lower bound $a + \delta \geq a/2$ follows from the $T_0 = \max(1, 1 + 2K_0/a)$ choice. (5) The ratio identity $(a + \varepsilon)/(a + \delta) - 1 = (\varepsilon - \delta)/(a + \delta)$ and the final division direction (larger denominator $\Rightarrow$ smaller quotient) are correct. (6) Docstrings do not overclaim. The reviewer independently confirmed all six.

4 Simplifications applied

Six: one dead open, five dead locals. open **Real** is dead (every `Real.sqrt/Real.pi/Real.mul_self_sqrt/Real.sqrt_pos` qualified); open **MeasureTheory** stays (bare $\int$ in the `set D/set N`). Then four unreferenced `set ...` with **h** equation names (`ha_def, hT0_def, hD_def, hN_def` — only the abbreviations are used) and the dead **have** `hsq` (the

fact $\sqrt{T} \cdot \sqrt{T} = T$, never referenced). Build green at 8256 jobs, warning set unchanged at zero, signatures byte-identical. GPT affirmed.

5 Disagreements and open questions

None. The reviewer’s case-split eliminations (in `hK0T_1e` and the `Tendsto` proof), the `show→change` suggestion on the $a + \delta = \sqrt{T} D$ rewrites, and the T_0 witness-weakening were declined as proof-golf — and note `change` cannot replace that rewrite, since $a + (\sqrt{T} D - a)$ is only *propositionally* (not definitionally) equal to $\sqrt{T} D$.

6 Lessons

- **Grep substring false-positives can hide a dead local — and reveal one.** The dead `hsq` appeared to have “11 occurrences” under a bare `grep hsq`, but ten were substring hits inside `hsqrt_T_pos/hsqrtTD_lb` etc.; only the definition line was a real occurrence, confirming it dead. Read the grep hits, don’t trust the count.
- **The unusedVariables linter silently misses set ... with h names and some haves.** As in the TailBound review, a silent linter is not proof a local is used; the rebuild (here, with a before/after warning-count comparison against `HEAD`) is the arbiter.
- **A multi-step analytic estimate deserves a per-claim fidelity ledger.** The six checks (constant, limit, rate, lower bound, ratio identity, division direction) are each a place the proof could have silently gone wrong; enumerating them is the review’s value over the green build.

7 Follow-ups

- Remaining `OneD/` surface: `HarmonicCovGlobalConstancy`, `HarmonicPerturbedMoments`, the `Harmonic*` monomials; and the `Multi/Covariance*` trio (sectioned).
- Proof-golf forward tide: collapse the two $K = 0$ / $K > 0$ case splits (both branches go through the general argument).